COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	9/4/26	**DATE SUBMITTED:**	9/4/26
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ACTIVITY NO. 7.1 Using Sorting Algorithms

1. PROCEDURE OUTPUT

What procedures did you perform?

SW7.1 > main.cpp > main()

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
#include "sortingAlgo.h"

const int maxSize = 100;

void displayArr(int arr[]);

int main(){

    //create an array with unsorted list with 100 maximum size
    int unsortedArr[maxSize];
    // use time (0) as seed for random values
    srand(time(0));
    //populate the array with random values
    for(int i = 0; i < maxSize; i++){
        unsortedArr[i] = rand()%100;
    }
    displayArr(unsortedArr);

    std::cout<<"\n Bubble Sort testing: \n";
    bubbleSort(unsortedArr, maxSize);
    displayArr(unsortedArr);

    std::cout<<"\n selectionSort testing: \n";
    selectionSort(unsortedArr, maxSize);
    displayArr(unsortedArr);

    std::cout<<"\n insertionSort testing: \n";
    insertionSort2(unsortedArr, maxSize);
```

```
30      std::cout<<"\n insertionSort testing: \n";
31      insertionSort2(unsortedArr, maxSize);
32      displayArr(unsortedArr);
33
34  }
35
36  void displayArr(int arr[]){
37      for (int i = 0; i < maxSize; i++) {
38          std::cout << arr[i] << " ";
39      }
40      std::cout << std::endl;
41  }
```

```cpp
#ifndef SORTINGALGO_H
#define SORTINGALGO_H
#include <iostream>

template<typename T>
void bubbleSort(T arr[], int maxSize){
    for(int i = 0; i < maxSize; i++){
        for(int j = i + 1; j < maxSize; j++){
            if(arr[i]>arr[j]){
                std::swap(arr[j],arr[i]);
            }
        }
    }
}

template<typename T>
int routineSmallest(T arr[], int K, int maxSize){
    int position;
    T smallestElem = arr[K];
    position = K;

    for(int j = K+1; j < maxSize; j++){
        if(arr[j] < smallestElem){
            smallestElem = arr[j];
            position = j;
        }
    }
    return position;
}

template<typename T>
```

```
30
31  template<typename T>
32  void selectionSort(T arr[], int maxSize){
33      int POS;
34      for(int i = 0; i < maxSize; i++){
35          POS=routineSmallest(arr, i, maxSize);
36          std::swap(arr[i], arr[POS]);
37      }
38  }
39
40  template <typename T>
41  void insertionSort(T arr[], int maxSize){
42      int k-1, j, temp;
43
44      for(int k=1; k<maxSize; k++){
45         temp = arr[k];
46  }
47
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following questions:

1. What is a sorting algorithm?
Is an arrangement type where it arranges the data into a specific order such as numbers or l, to make information neat and easier to search

2.Where can sorting algorithms be used?
In databases, finances, search engines, and organizing files in systems.

3. Explain the different types of sorting algorithms.

Bubble Sort:
Compares adjacent pairs of elements from left to right and swaps them if they are in the wrong order.

Selection Sort:
Divides the array into a sorted and unsorted section, then it will sssssrepeatedlys scan the unsorted section to find the minimum element, then swaps it with the first element of the unsorted section.

Insertion Sort:
It takes an element from the unsorted position and shifts larger elements in the sorted position to the right to insert it into its correct position.

Merge Sort:
It splits it repeatedly in half until individual elements remain.

Quick Sort:
Picks a single element and partitions the array so that all smaller elements move to its left and all larger elements move to its right.

4. Give sample programs in C++ that uses sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.

Bubble:

SW7.1 > Types of sorting > Bubble > bubbleSort(int [], int)

```cpp
void bubbleSort(int arr[], int size) {
    for (int i = 0; i < size - 1; i++) {
        for (int j = 0; j < size - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                std::swap(arr[j], arr[j + 1]);
            }
        }
    }
}
```

Compares adjacent pairs of elements from left to right and swaps them if they are in the wrong order.

Insertion:

```cpp
void insertionSort(int arr[], int size) {
    for (int i = 1; i < size; i++) {
        int key = arr[i], j = i - 1;
        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
}
```

It takes an element from the unsorted position and shifts larger elements in the sorted position to the right to insert it into its correct position.

Selection:

```
void selectionSort(int arr[], int size) {
    for (int i = 0; i < size - 1; i++) {
        int minIdx = i;
        for (int j = i + 1; j < size; j++) {
            if (arr[j] < arr[minIdx]) minIdx = j;
        }
        std::swap(arr[i], arr[minIdx]);
    }
}
```

Divides the array into a sorted and unsorted section, then it will sssssrepeatedlys scan the unsorted section to find the minimum element, then swaps it with the first element of the unsorted section.

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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